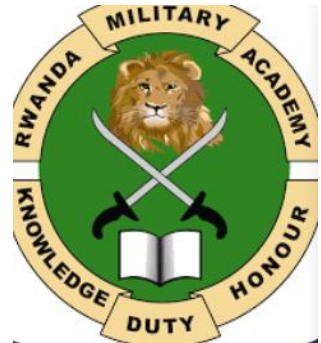




UNIVERSITY OF RWANDA-GAKO CAMPUS

COLLEGE OF SCIENCE AND TECHNOLOGY-SCHOOL OF ICT

MOBILE APPLICATION SYSTEM AND DESIGN

On 29th March 2026

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COMPUTER ENGINEERING

YEAR 3

Q1. Dependencies Used and reasons why Used? And why SQLite is necessary to store data locally.

> SQLite : is used to work with SQLite database for storing data locally on the device

> Path : is used to define and manage the database file location.

SQLite is necessary because it allows the application to store and manage data offline without requiring internet access.

Q2. Handling database exceptions

Database exceptions can occur due to issues like uninitialized database, failed operations, or invalid data.

> Database not initialized

`OpenDatabase()` inside an `initDatabase()` function can be used and wrapped in try-catch to show a friendly error message if initialization fails.

> Insert/Update/Delete errors

Example, Inserting a post with missing fields.

Handle with try-catch and return a `future<bool>` to indicate success/failure.

> Invalid or Corrupted data

Input validation (Validate inputs before saving. e.g., title not empty) is also used to handle corrupted or invalid data.

If corruption occurs, deletion and recreation of the database file can also be done.

Q3. SQLite in flutter explained

> Database vs Table

A database is a container that stores data, while a table is a structure inside the database consisting of rows and columns (e.g., posts table with id, title, content).

> CRUD Operations

Create: Inserting new data into the table

```
db.insert('posts', postMap)
```

Read: Retrieving data from the table

```
db.query('posts') or
```

```
db.rawQuery("SELECT * FROM posts")
```

Update: Modifying existing data

```
db.update('posts', postMap, where: 'id=?', whereArgs: [id])
```

Delete: Removing data from the table

```
db.delete('posts', where: 'id=?', whereArgs: [id])
```

> Asynchronous Interaction

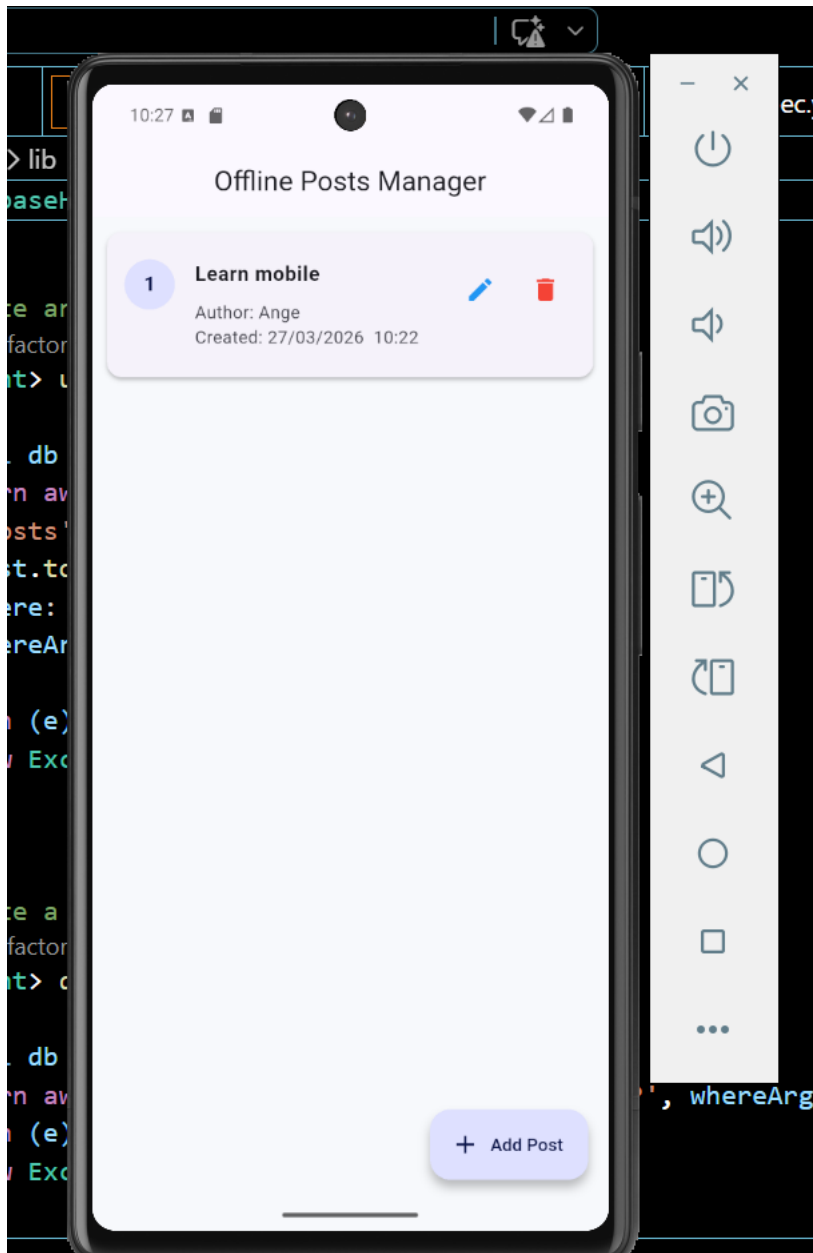
Flutter interacts with SQLite asynchronously

Using Future, which means database operations happen in the background without blocking the UI.

and it uses Future and async/await for database calls.

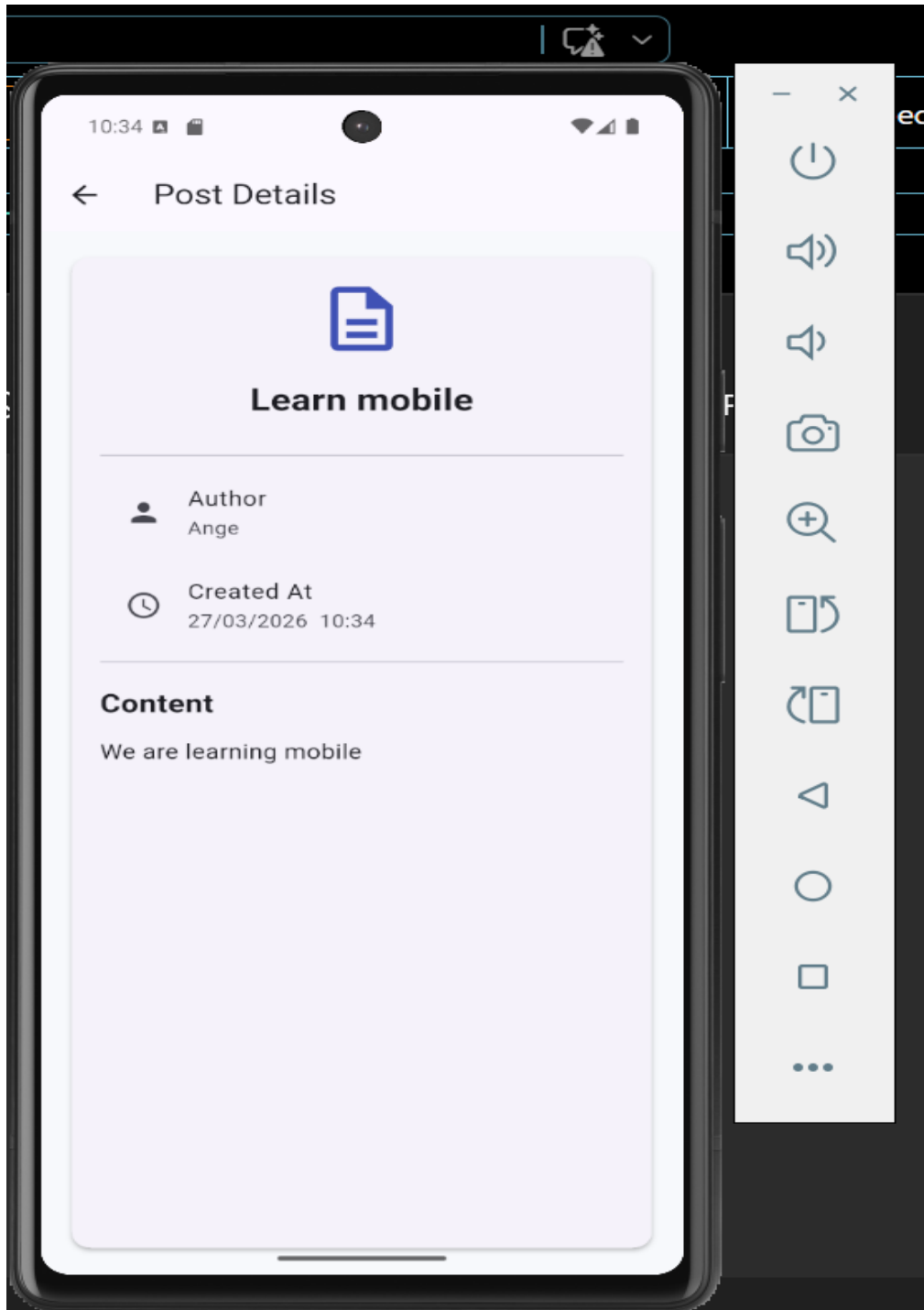
1. HOME SCREEN (View posts)

This screen displays all posts stored locally in the SQLite database. The data is retrieved using **Future** and displayed using **ListView**. Each post shows its title and body, allowing users to view stored information offline.



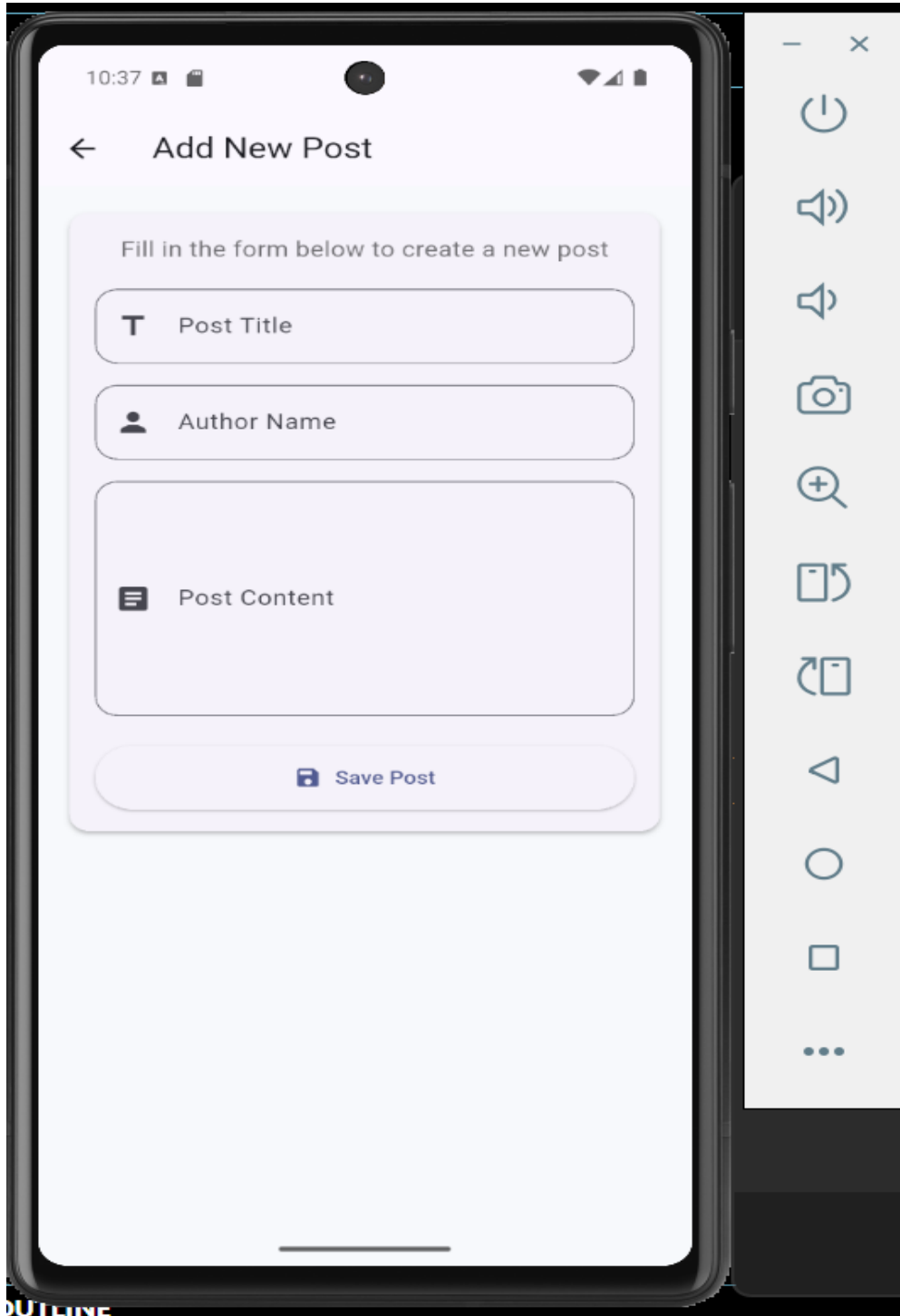
2. READ THE DETAILS OF A POST

This screen displays the full details of a selected post. Allowing the user to read the full details directly from the list view.



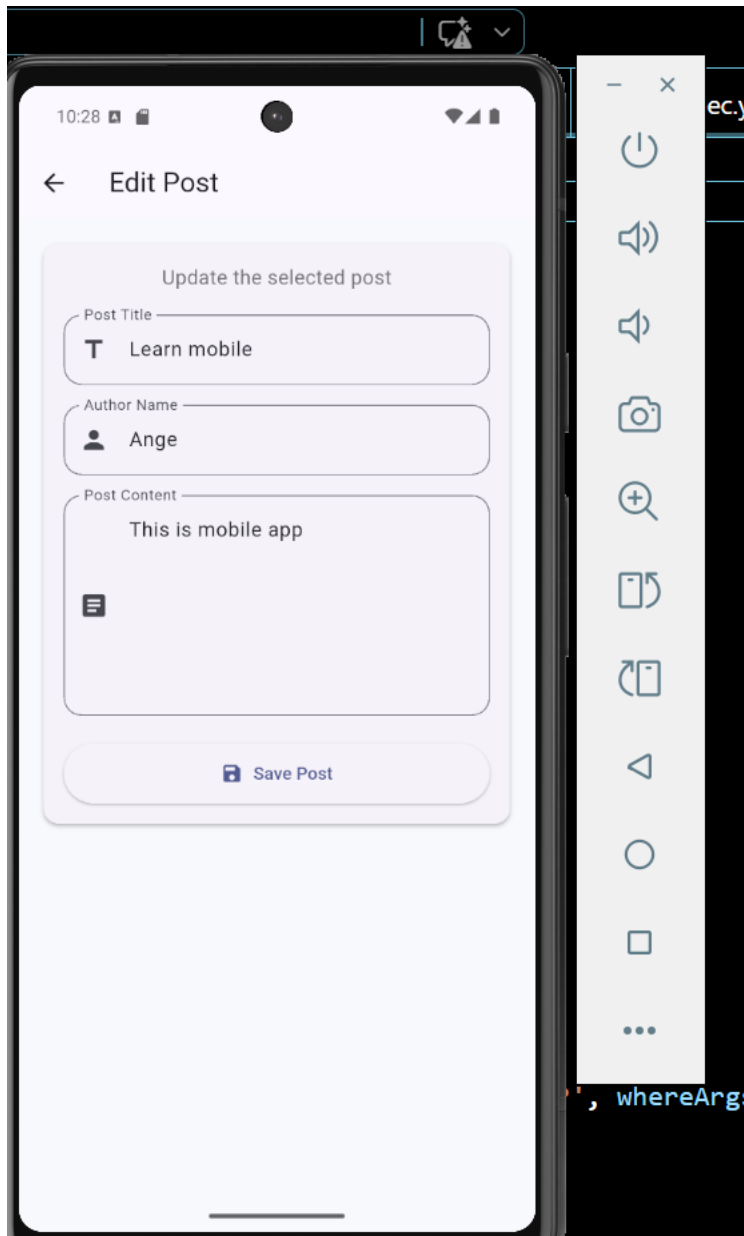
3. ADDING A NEW POST

Here, the user creates a new post by entering a title and body. When the user clicks the save button, the data is inserted into the SQLite database using the insert operation.



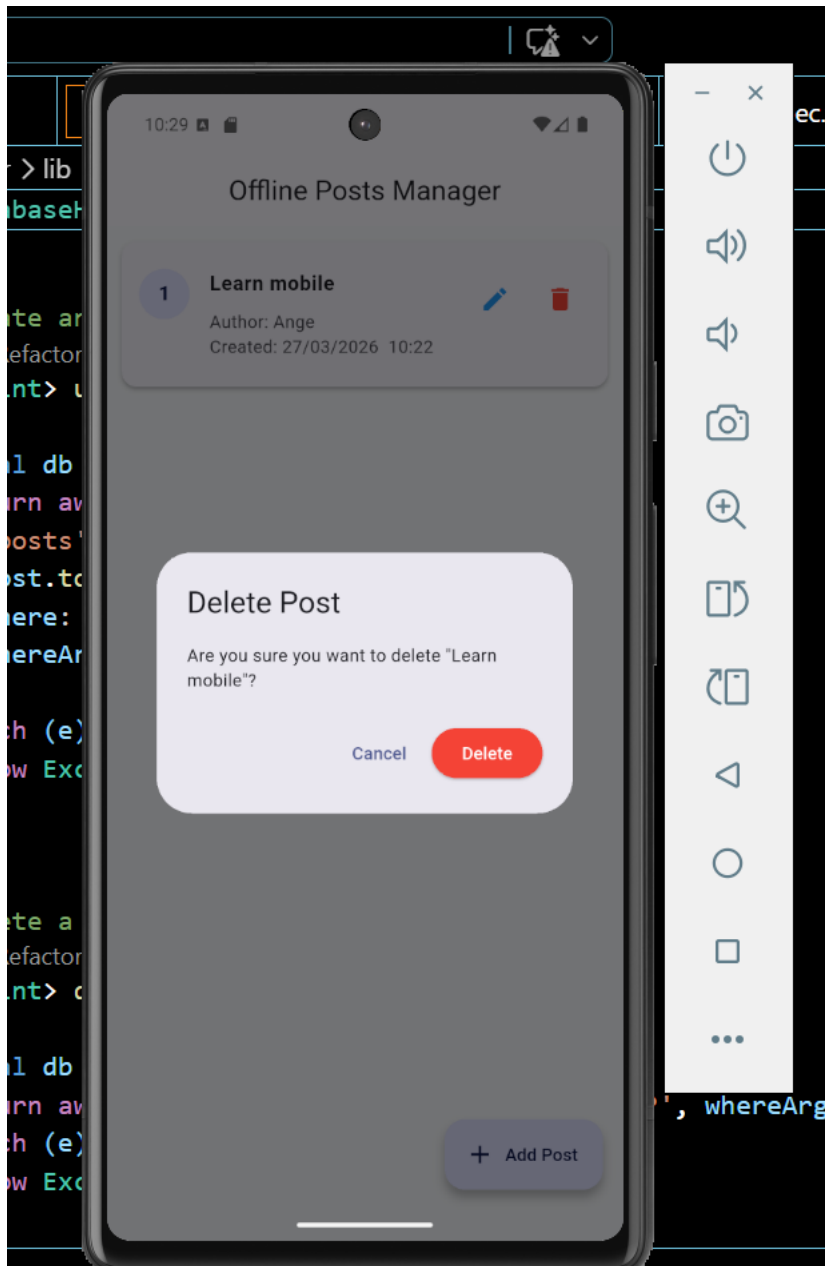
4. EDITING AN EXISTING POST

This screen is used to update an existing post. The fields are pre-filled with the selected post data, and the user can modify the content. The updated data is saved using the update operation in SQLite.

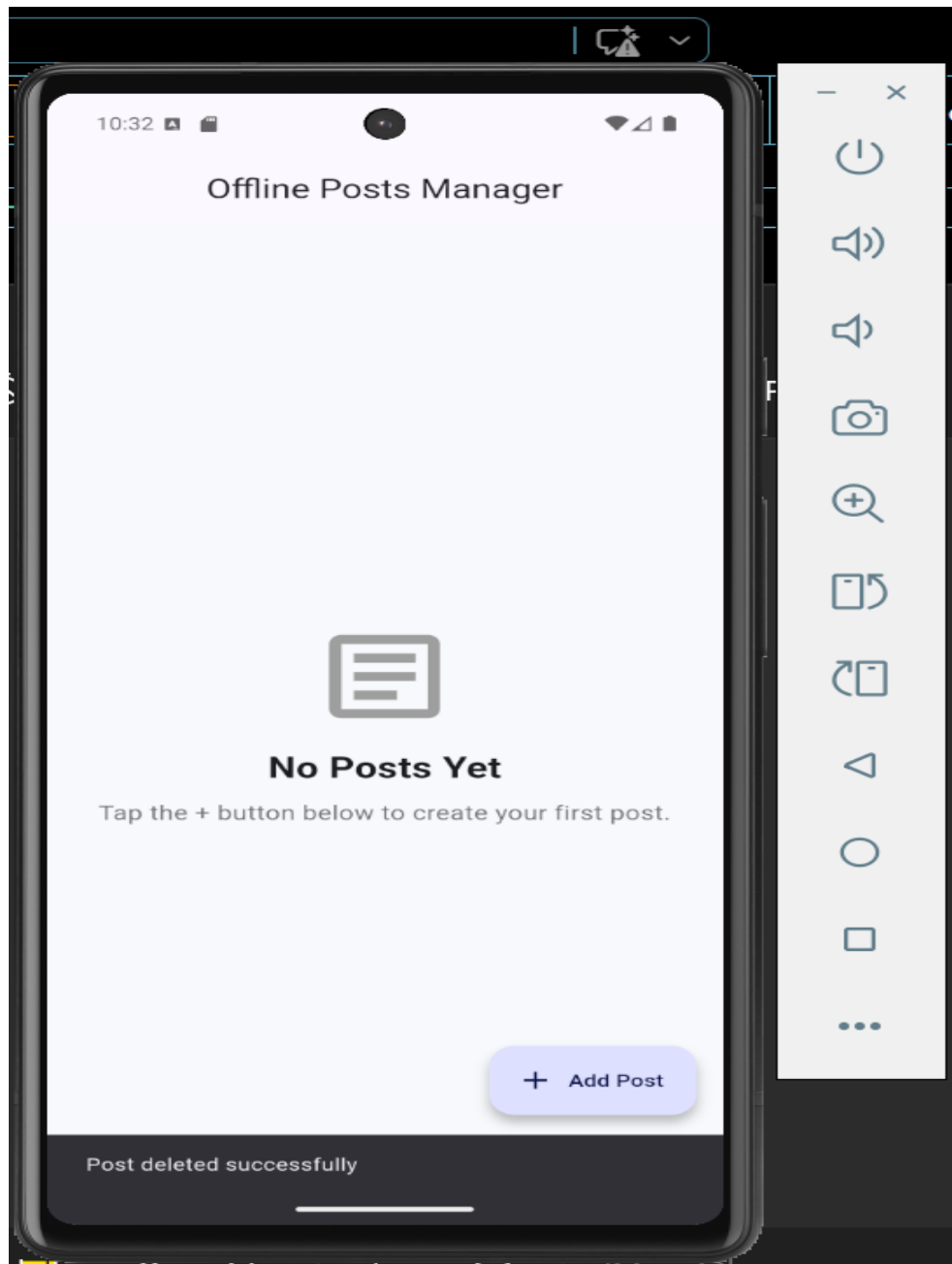


5. DELETING A POST

This action allows the user to delete a post from the database by clicking the delete icon. The selected post is removed permanently using the delete operation, and the list updates automatically.



After deleting a post, message” Post deleted successfully” displayed.



CONCLUSION

This application demonstrates how Flutter can use SQLite to store and manage data locally. It supports full CRUD operations and works without internet connection.